

Package ‘CrossingTools’

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Title Tools for Cross Optimization in Plant Breeding Programs

Version 1.1.0

Description Mate allocation is a critical component of plant breeding programs, aiming to generate superior progenies and improved populations over time. While numerous methodologies for mate allocation exist, few software tools offer a unified and scalable framework that integrates these methodologies in a user-friendly manner. 'CrossingTools' addresses this gap by providing a flexible, efficient solution for optimizing mating decisions across a wide range of breeding schemes. The package supports genomic BLUP and selection indices for various variance structures and implements multiple selection criteria to evaluate cross potential, including expected mean, optimal haploid value (OHV), and superior progeny value (SPV). Different methods are provided for calculating additive and dominance segregation variances for both inbred and heterozygous parents, making 'CrossingTools' suitable for a wide range of breeding strategies. After criterion calculation, users can rank potential crosses or apply the built-in optimal crossing selection routine, which balances genetic gain and diversity to support long-term improvement. The package combines R's scripting flexibility with high-performance C++ routines via 'Rcpp' and 'RcppArmadillo', ensuring scalability for large populations with dense marker data.

License GPL-3

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backsolve_marker_effects

Backsolve marker effects from individual effects

Description

Backsolves marker effects from individual genetic effects using

$$\mu = s\mathbf{M}^T\mathbf{G}^{-1}\mathbf{g},$$

where $\mathbf{M}$ is the marker matrix, $\mathbf{G}$ is the genomic relationship matrix, $\mathbf{g}$ contains individual effects, and s is a scaling factor determined internally.

Usage

```
backsolve_marker_effects(marker.mat, G.mat, genotype.effects)
```

Arguments

<code>marker.mat</code>	Numeric matrix with individuals in rows and markers in columns.
<code>G.mat</code>	Numeric square matrix containing genomic relationships among the individuals in <code>marker.mat</code> .
<code>genotype.effects</code>	Numeric vector, matrix, or data frame containing individual genetic effects. Rows must correspond to individuals in <code>marker.mat</code> ; columns represent traits.

Details

Multiple traits can be processed simultaneously by supplying `genotype.effects` as a matrix or data frame with one column per trait.

Value

A data frame with markers in rows and traits in columns containing the backsolved marker effects.

`calc_midparent_inbred` *Calculate predicted additive means for specified crosses*

Description

Calculates the expected genomic breeding value of proposed two-way or four-way crosses as the mean of the parental genomic breeding values.

Usage

```
calc_midparent_inbred(
  crosses,
  marker.mat,
  marker.effects,
  weights = NULL,
  nthreads = 4L
)
```

Arguments

<code>crosses</code>	Matrix or data frame with two columns for two-way crosses or four columns for four-way crosses. Parent identifiers may be row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> .
<code>marker.mat</code>	Numeric marker dosage matrix with individuals in rows and markers in columns, coded 0, 1, and 2 for the counted allele.
<code>marker.effects</code>	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects must be parameterised relative to the reference population represented by <code>marker.mat</code> . Its number of rows must equal <code>ncol(marker.mat)</code> .
<code>weights</code>	Optional numeric vector with one weight per trait. When supplied, a weighted genomic breeding value index is calculated for each cross.
<code>nthreads</code>	Positive integer. Number of computational threads.

Details

Multiple traits can be evaluated simultaneously. Optional trait weights can be used to calculate a weighted genomic breeding value index for each cross.

Value

If `weights = NULL`, a data frame containing the parental identifiers followed by one GEBV.<trait> column per trait.

If `weights` is supplied, a list containing:

`cross.df` The parental identifiers and trait-specific genomic breeding values.

`index.df` The parental identifiers and the weighted index GEBV.IDX.

calc_midparent_outcross

Calculate predicted means for two-way outcrosses

Description

Calculates the expected genomic breeding value and expected total genetic value of proposed two-way crosses from additive and dominance marker effects.

Usage

```
calc_midparent_outcross(
  crosses,
  marker.mat,
  marker.effects.A,
  marker.effects.D = NULL,
  weights = NULL,
  nthreads = 4L
)
```

Arguments

- | | |
|-------------------------------|--|
| <code>crosses</code> | Matrix or data frame with two columns specifying the parents of each proposed cross. Parent identifiers may be row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> . |
| <code>marker.mat</code> | Numeric marker dosage matrix with individuals in rows and markers in columns. |
| <code>marker.effects.A</code> | Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects must be parameterised relative to the reference population represented by <code>hap.mat1</code> and <code>hap.mat2</code> . Its number of rows must equal <code>ncol(hap.mat1)</code> . |
| <code>marker.effects.D</code> | Numeric matrix of dominance effects with markers in rows and traits in columns, used with the statistical dominance-deviation parameterisation. It must have the same dimensions as <code>marker.effects.A</code> . |
| <code>weights</code> | Optional numeric vector with one weight per trait. When supplied, weighted indices are calculated from the trait-specific GEBVs and TGVs. |
| <code>nthreads</code> | Positive integer. Number of computational threads. |

Details

The expected genomic breeding value is the mid-parent additive value. The expected total genetic value additionally includes the expected dominance contribution of the F1. If dominance effects are not supplied, they are assumed to be zero.

Multiple traits can be evaluated simultaneously. Optional trait weights can be used to calculate additive and total-genetic-value indices.

Value

If `weights = NULL`, a data frame containing the parental identifiers followed by `GEBV.<trait>` and `TGV.<trait>` for each trait.

If `weights` is supplied, a list containing:

`cross.df` The parental identifiers and trait-specific GEBVs and TGVs.

`index.df` The parental identifiers and the weighted indices `GEBV.IDX` and `TGV.IDX`.

calc_optimal_haploid_value

Calculates the Optimal Haploid Value (OHV) of proposed two-way or four-way crosses from marker effects. If haplotype blocks are provided, marker effects are first aggregated within blocks; otherwise, each marker is treated as an independent block. Trait-specific OHVs and an optional weighted OHV index are returned.

Description

Calculates the Optimal Haploid Value (OHV) of proposed two-way or four-way crosses from marker effects. If haplotype blocks are provided, marker effects are first aggregated within blocks; otherwise, each marker is treated as an independent block. Trait-specific OHVs and an optional weighted OHV index are returned.

Usage

```
calc_optimal_haploid_value(
  crosses,
  marker.mat,
  marker.effects,
  weights = NULL,
  haplotype.blocks = NULL,
  nthreads = 4L
)
```

Arguments

<code>crosses</code>	Matrix or data frame with two columns for two-way crosses or four columns for four-way crosses. Parent identifiers may be integer row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> .
<code>marker.mat</code>	Numeric marker matrix with individuals in rows and markers in columns.
<code>marker.effects</code>	Numeric matrix of marker effects with markers in rows and traits in columns. Its number of rows must equal <code>ncol(marker.mat)</code> .

weights	Optional numeric vector with one weight per trait. When supplied, a weighted OHV index is calculated as the linear combination of the trait-specific OHVs.
haplotype.blocks	Optional data frame defining the haplotype blocks. It must contain: - block Block identifier. Markers with the same identifier belong to the same haplotype block. - site Marker identifier given as a column index of marker.mat or as a marker name matching colnames(marker.mat). Each marker may occur in at most one block. If NULL, each marker is treated as an independent block.
nthreads	Positive integer. Number of computational threads.

Value

If weights = NULL, a data frame containing the original cross definitions followed by one OHV.<trait> column per trait.

If weights is supplied, a list containing:

cross.df The original cross definitions and the trait-specific OHVs.

index.df The cross definitions and the weighted OHV index in column OHV.IDX.

calc_spv_inbred	<i>Calculate segregation variance and superior progeny values for inbred lines</i>
-----------------	--

Description

Calculates expected genomic breeding values, segregation variances, and superior progeny values for doubled haploid or recombinant inbred line families derived from proposed two-way or four-way crosses.

Usage

```
calc_spv_inbred(
  crosses,
  genetic.map,
  marker.mat,
  marker.effects,
  t,
  intensity = 1,
  type = "DH",
  weights = NULL,
  covariance = FALSE,
  method = 1,
  nthreads = 4L
)
```

Arguments

<code>crosses</code>	Matrix or data frame with two columns for two-way crosses or four columns for four-way crosses. Parent identifiers may be row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> .
<code>genetic.map</code>	Data frame describing the genetic map. It must contain: <code>site</code> Marker index or marker name matching <code>colnames(marker.mat)</code> . <code>chr</code> Chromosome identifier. <code>pos</code> Marker position on the chromosome in Morgan. All markers in <code>marker.mat</code> must be represented.
<code>marker.mat</code>	Numeric marker dosage matrix with individuals in rows and markers in columns, coded 0, 1, and 2 for the counted allele.
<code>marker.effects</code>	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects must be parameterised relative to the reference population represented by <code>marker.mat</code> . Its number of rows must equal <code>ncol(marker.mat)</code> .
<code>t</code>	Integer. Number of random-mating generations before doubled haploid or recombinant inbred line development.
<code>intensity</code>	Numeric scalar giving the standardized selection intensity used to calculate superior progeny values. The default is 1.
<code>type</code>	Character. Offspring type, either "DH" for doubled haploid lines or "RIL" for recombinant inbred lines.
<code>weights</code>	Optional numeric vector with one weight per trait. When supplied, weighted index values are calculated for each cross.
<code>covariance</code>	Logical. If TRUE, also calculate segregation covariance matrices among traits for each cross.
<code>method</code>	Method used to calculate segregation variance. Either 1 or "lehermeier" for the Lehermeier method, or 2 or "osthushenrich" for the Osthushenrich method. The latter is currently available only for two-way crosses.
<code>nthreads</code>	Positive integer. Number of computational threads.

Details

The calculation can account for additional generations of random mating before line development. Multiple traits and optional weighted trait indices are supported.

Value

If neither `weights` nor trait covariances are requested, a data frame containing the parental identifiers and, for each trait, `GEBV.<trait>`, `var.<trait>`, and `SPV.<trait>`.

If `weights` is supplied, a list additionally containing `index.df`, with columns `GEBV.IDX`, `var.IDX`, and `SPV.IDX`.

If `covariance = TRUE`, a list containing:

`cross.df` Cross-specific trait values and segregation variances.

`index.df` Weighted index values, if `weights` is supplied.

`covariances` Segregation covariance matrices for each cross.

calc_spv_outcross	<i>Calculate segregation variance and superior progeny values for outcrosses</i>
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Description

Calculates expected genomic breeding values, expected total genetic values, additive and dominance segregation variances, and superior progeny values for proposed F1 crosses.

Usage

```
calc_spv_outcross(
  crosses,
  genetic.map,
  hap.mat1,
  hap.mat2,
  marker.effects.A,
  marker.effects.D,
  intensity = 1,
  weights = NULL,
  covariance = FALSE,
  nthreads = 4L
)
```

Arguments

crosses	Matrix or data frame with two columns specifying the parents of each proposed cross. Parent identifiers may be row indices of hap.mat1 and hap.mat2, or character identifiers matching their row names.
genetic.map	Data frame describing the genetic map. It must contain: site Marker identifier given as a column index or as a marker name matching colnames(hap.mat1). chr Chromosome identifier. pos Marker position on the chromosome in Morgan. All markers in the haplotype matrices must be represented.
hap.mat1	Numeric haplotype matrix with individuals in rows and markers in columns, containing the first haplotype of each individual.
hap.mat2	Numeric haplotype matrix with the same dimensions as hap.mat1, containing the second haplotype of each individual.
marker.effects.A	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects must be parameterised relative to the reference population represented by hap.mat1 and hap.mat2. Its number of rows must equal ncol(hap.mat1).
marker.effects.D	Numeric matrix of dominance effects with markers in rows and traits in columns, used with the statistical dominance-deviation parameterisation. It must have the same dimensions as marker.effects.A.

intensity	Numeric scalar giving the standardized selection intensity used to calculate SPV and TSPV. The default is 1.
weights	Optional numeric vector with one weight per trait. When supplied, weighted index values are calculated for each cross.
covariance	Logical. If TRUE, also calculate breeding-value, dominance-deviation, and combined additive–dominance segregation covariance matrices among traits for each cross.
nthreads	Positive integer. Number of computational threads.

Details

Additive superior progeny values are calculated from the breeding-value segregation variance, whereas total superior progeny values are calculated from the total genetic segregation variance, including breeding-value variance, dominance-deviation variance, and their covariance.

Value

If neither weights nor trait covariances are requested, a data frame containing the parental identifiers and, for each trait, GEBV.<trait>, TGV.<trait>, var.A.<trait>, SPV.<trait>, var.D.<trait>, var.TGV.<trait>, and TSPV.<trait>.

If weights is supplied, index.df contains GEBV.IDX, TGV.IDX, var.A.IDX, SPV.IDX, var.D.IDX, var.TGV.IDX, and TSPV.IDX. The total genetic segregation variance var.TGV.IDX includes breeding-value variance, dominance-deviation variance, and their covariance.

If covariance = TRUE, a list containing:

cross.df Cross-specific trait values and segregation variances.

index.df Weighted index values, if weights is supplied.

additive.covariances Breeding-value segregation covariance matrices for each cross.

dominance.covariances Dominance-deviation segregation covariance matrices for each cross.

additive.dominance.covariances Combined additive–dominance covariance matrices, $\Sigma_{AD} + \Sigma_{DA}$, for each cross.

make_cross_plan

Create a crossing plan from candidate parents

Description

Generates all possible crosses among a set of candidate parents.

Usage

```
make_cross_plan(
  parents = NULL,
  self = FALSE,
  male.parents = NULL,
  female.parents = NULL
)
```

Arguments

parents	Optional vector of parent identifiers for an unsexed crossing design. At least two unique identifiers must be supplied.
self	Logical. If TRUE, include self-crosses when parents is supplied. Ignored for sexed crossing designs.
male.parents	Optional vector of male parent identifiers for a sexed crossing design.
female.parents	Optional vector of female parent identifiers for a sexed crossing design.

Details

For an unsexed design, all unique pairwise crosses are returned, optionally including self-crosses. For a sexed design, all combinations between the supplied male and female parents are returned.

Value

A data frame with one row per cross. For unsexed designs, the columns are parent1 and parent2. For sexed designs, the columns are male and female.

make_index	<i>Calculate a multi-trait selection index</i>
------------	--

Description

Calculates a selection index from multi-trait genomic predicted effects using either a desired-gains index or a Smith-Hazel index.

Usage

```
make_index(
  genotype.effects,
  weights = NULL,
  var.mat = NULL,
  desired.gains = NULL,
  nthreads = 4L
)
```

Arguments

genotype.effects	Numeric matrix or data frame containing genotype effects, with genotypes in rows and traits in columns.
weights	Optional numeric vector with one weight per trait. When supplied, a Smith-Hazel index is calculated as a weighted sum of the trait effects.
var.mat	Optional covariance matrix used for the desired-gains index. It may be either a trait covariance matrix or the posterior covariance matrix of the stacked genotype effects. If NULL, the trait covariance matrix is estimated from genotype.effects.
desired.gains	Optional numeric vector specifying the desired gain for each trait.
nthreads	Positive integer. Number of computational threads.

Details

For the desired-gains index, trait weights are derived from the specified desired gains and a trait covariance matrix. For the Smith-Hazel index, the supplied trait weights are used directly. Exactly one of `desired.gains` or `weights` must be provided.

Value

A list containing:

`index` A data frame containing the selection index value for each genotype.

`weights` The trait weights used to construct the index.

<code>optimize_cross_plan</code>	<i>Optimize a crossing plan using a genetic algorithm</i>
----------------------------------	---

Description

Selects a crossing plan of `ncrosses` parent pairs by maximizing a user-defined cross criterion while minimizing genomic similarity, or another relationship measure, derived from `G.mat`. Fixed crosses and upper and lower parent contribution constraints can be included.

Usage

```
optimize_cross_plan(
  candidate.crosses,
  fixed.crosses = NULL,
  ncrosses,
  target.angle = 0,
  criterion,
  criterion.fixed = NULL,
  G.mat,
  parents.upper = NA,
  parents.lower = NA,
  method = "pareto",
  return.params = FALSE,
  plot = TRUE,
  params = list(),
  nthreads = 4L
)
```

Arguments

<code>candidate.crosses</code>	Integer or character matrix or data frame with two columns. Candidate crosses from which the variable part of the crossing plan is selected. Self-crosses are not allowed.
<code>fixed.crosses</code>	Integer or character matrix or data frame with two columns, or <code>NULL</code> . Crosses included in every returned plan. These must use the same type of parent identifiers as <code>candidate.crosses</code> . Self-crosses are not allowed.
<code>ncrosses</code>	Integer. Total number of crosses in each returned plan, including fixed crosses. Must be at least <code>nrow(fixed.crosses)</code> .

target.angle	Numeric scalar between 0 and 90. Target trade-off angle used when method = "angle". An angle of 0 prioritizes the cross criterion, whereas 90 prioritizes minimum similarity.
criterion	Numeric vector with one cross-criterion value for each row of candidate.crosses.
criterion.fixed	Numeric vector with one cross-criterion value for each row of fixed.crosses, or NULL when no fixed crosses are used.
G.mat	Numeric square matrix containing genomic relationships, similarities, or another measure used to quantify parental contributions. Character parent identifiers require matching row names.
parents.upper	Numeric vector with one value per parent in G.mat, giving the maximum number of times each parent may occur across the complete crossing plan. Each cross contributes one occurrence to each parent. Values must be non-negative whole numbers or Inf. The default is Inf for every parent.
parents.lower	Numeric vector with one value per parent in G.mat, giving the minimum number of times each parent must occur across the complete crossing plan. Each cross contributes one occurrence to each parent. Values must be non-negative whole numbers and cannot exceed the corresponding values in parents.upper. The default is zero for every parent.
method	Character string specifying the optimization method. Must be either "angle" or "pareto".
return.params	Logical. Whether to return optimization diagnostics in addition to the selected crossing plan or Pareto solutions.
plot	Logical. Whether to display the estimated Pareto frontier when method = "pareto". Ignored when method = "angle".
params	Named list of genetic-algorithm parameters. Unspecified entries use their default values: probability.mutate Numeric scalar in $[0, 1]$. Mutation probability for each offspring. n.mutate Non-negative integer. Number of positions changed when mutation occurs. n.select Integer of at least 2. Number of solutions selected in each generation. n.pop Integer of at least 2. Population size. max.generation Positive integer. Maximum number of generations. max.iteration Positive integer. Number of generations without improvement after which optimization stops. angle.penalty Non-negative numeric scalar. Penalty for deviation from target.angle, used only when method = "angle".
nthreads	Positive integer. Number of computational threads.

Details

Two optimization methods are available:

method = "angle" Returns one crossing plan targeting the trade-off specified by target.angle. An angle of 0 degrees prioritizes consequently, greater diversity.

method = "pareto" Returns an approximation of the Pareto frontier containing plans with different trade-offs between the cross criterion and similarity. This method is generally more computationally demanding.

Value

For method = "angle" and return.params = FALSE, a data frame with columns parent1 and parent2 describing the selected crossing plan.

For method = "angle" and return.params = TRUE, a list containing:

crossPlan The selected crossing plan as a data frame with columns parent1 and parent2.

uMax, simMax The maximum attainable cross criterion and the corresponding similarity.

uBest, simBest The cross criterion and similarity of the selected solution.

angleBest, lenBest The angle and length of the selected solution in normalized objective space.

For method = "pareto", a list containing:

pareto.plans A list of pareto crossing plans, each represented by a data frame with columns parent1 and parent2.

pareto.frontier A data frame containing the solution identifier pareto.id, cross criterion u, and similarity sim.

plot_cross_plan	<i>Plot predicted cross distributions by trait</i>
-----------------	--

Description

Creates trait-specific ridgeline plots of the predicted distributions of crosses in a crossing plan. Distributions are generated from the available cross means and segregation variances.

Usage

```
plot_cross_plan(cross.plan, cross.df, traits = NULL, nsamples = 1000L)
```

Arguments

cross.plan	Data frame, matrix, or object coercible to a data frame containing the crossing plan. It must have two columns for two-way crosses or four columns for four-way crosses. Rows determine the displayed cross labels and order.
cross.df	Data frame, matrix, or object coercible to a data frame containing cross-specific predictions, typically produced by a variance or optimal-haploid-value function. Trait-specific columns must follow the naming convention <metric>.<trait>, for example GEBV.Yield, SPV.Yield, var.A.Yield, or var.D.Yield. Crosses are matched using columns parent1 and parent2, male and female, or the first two columns. Four-way crosses require columns parent1 to parent4.
traits	Character vector specifying the trait names to plot, or NULL to plot all traits detected in cross.df.
nsamples	Positive integer. Number of Monte Carlo samples generated per cross, trait, and distribution type.

Details

Additive and, when available, additive-plus-dominance distributions are shown separately. Available cross criteria, including GEBV, TGV, SPV, TSPV, and OHV, are added as points. A dotted vertical line indicates the mean GEBV across crosses for each trait.

Value

A ggplot object containing trait-specific ridgeline distributions and available point estimates.

predict_response	<i>Predict the response to multi-trait index selection</i>
------------------	--

Description

Summarizes a linear multi-trait selection index from a trait covariance matrix and either desired gains or fixed trait weights.

Usage

```
predict_response(
  var.mat = NULL,
  desired.gains = NULL,
  weights = NULL,
  intensity = 1,
  plot = TRUE
)
```

Arguments

var.mat	Numeric covariance matrix. It may be either a trait covariance matrix or a block-structured covariance matrix containing one trait covariance block per genotype. In the latter case, the trait covariance matrix is obtained by averaging the diagonal blocks.
desired.gains	Optional numeric vector specifying the desired gain for each trait.
weights	Optional numeric vector specifying the index weight for each trait.
intensity	Numeric scalar giving the standardized selection intensity used to scale the expected responses. The default is 1.
plot	Logical. If TRUE, plot the expected response for each trait. The default is TRUE.

Details

For a desired-gains index, trait weights are derived from the supplied desired gains. For an index with fixed weights, the supplied weights are used directly. Exactly one of `desired.gains` or `weights` must be provided.

The function returns the index variance, the expected response in the index, and the expected response and correlation with the index for each trait.

Value

A list containing:

`overall.df` A data frame containing the index variance `index.var` and expected index response `gain.index`.

`trait.df` A data frame containing, for each trait, its variance, correlation with the index, expected response, and index weight.

summarize_cross_plan *Summarize the expected performance of a crossing plan*

Description

Calculates the mean predicted performance of the crosses in `cross.plan` for each available trait and value metric, including GEBV, TGV, SPV, TSPV, and OHV. Crosses are matched to `cross.df` using their parental identifiers.

Usage

```
summarize_cross_plan(
  cross.plan,
  cross.df,
  plot = T,
  reference.pop = NA,
  weights = NA
)
```

Arguments

<code>cross.plan</code>	Data frame, matrix, or object coercible to a data frame containing the crossing plan. It must have two columns for two-way crosses or four columns for four-way crosses.
<code>cross.df</code>	Data frame, matrix, or object coercible to a data frame containing predictions for the candidate crosses. Trait-specific value columns must follow the naming convention <code><metric>.<trait></code> , for example <code>GEBV.Yield</code> or <code>SPV.Height</code> . Crosses are matched using columns <code>parent1</code> and <code>parent2</code> , male and female, or the first two columns. Four-way crosses require columns <code>parent1</code> to <code>parent4</code> .
<code>plot</code>	Logical. If TRUE, plot the predicted mean GEBV response.
<code>reference.pop</code>	Optional numeric matrix or data frame with individuals in rows and traits in columns. When supplied, predicted responses are expressed relative to the corresponding trait means. It must contain at least two individuals and all traits represented in <code>cross.df</code> .
<code>weights</code>	Optional numeric vector of trait weights, in the same order as the traits in the returned summary. When supplied together with <code>reference.pop</code> , it is used to calculate and display the desired response direction.

Details

Optionally, the mean GEBV response can be plotted. When `reference.pop` is supplied, responses are expressed relative to the reference-population means. When trait weights are also supplied, the corresponding desired-response direction is shown for comparison.

Value

A data frame with one row per trait. The output contains:

`trait` Trait identifier parsed from the prediction-column names.
`ncrosses` Number of crosses in `cross.plan`.

mean.GEBV, mean.TGV, mean.SPV, mean.TSPV, mean.OHV Mean value of each available metric across the selected crosses. Metrics not available for a particular trait are returned as NA.

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